Circles

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

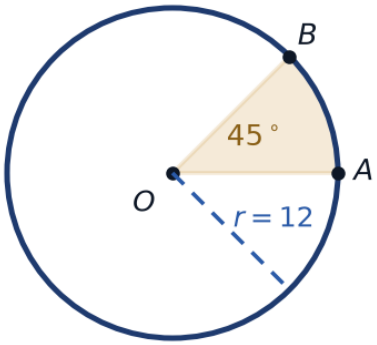
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC studying inequalities in one triangle, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC studying the triangle inequality, $m\angle A = 44^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given $f(x) = x^2 - 4x + 7$ with restricted domain $x \geq 2$, find an explicit formula for $f^{-1}(x)$ and state its domain.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An investment fund gains 30% in value during Year 1, decreases by 20% in Year 2, and increases by 10% in Year 3. Determine the overall net percent change of the fund across the three-year period.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 + 2x - 8 = 0$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A fair coin is tossed twice. Find $P(\text{two heads})$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A school theater sold 250 tickets for a play, raising a total of $1,750$. $Adulttickets cost$9 each and student tickets cost 5 each. How many adult tickets were sold?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A recipe is scaled from 2 to 5 servings. How many cups replace 4 cups?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Determine the equation of the circle in the xy-plane for which the segment connecting $A(2,\ -3)$ and $B(8, 5)$ serves as a diameter.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedWhat is the set of all real solutions to the radical equation $\sqrt{2x-5} = x-4$?

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Convert the quadratic function $y = 2x^2 - 12x + 11$ into vertex form $y = a(x-h)^2 + k$. State the coordinates of the vertex and the line of symmetry.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains 5 red marbles, 4 blue marbles, and 3 green marbles. Two marbles are drawn at random one after another without replacement. What is the probability that both marbles are red?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The scores on a statewide standardized test are normally distributed with a mean of $\mu = 500$ and a standard deviation of $\sigma = 100$. Approximately what percentage of test takers score between 400 and 700?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability

A fair die is rolled twice. Find $P(\text{both show } 6)$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Lines & Angles

A square has area 36. Find its perimeter.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Triangles & Congruence

Suppose y varies directly as x^2 and inversely as z. If $y = 18$ when $x = 3$ and $z = 4$, find the value of y when $x = 5$ and $z = 10$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A car travels at 60 mph for 2.5 hours. Find the distance.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A jacket costs USD 50 after a 20% discount. Find the original price.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

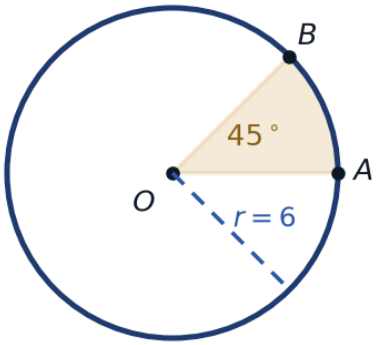
Solve the equation $5e^{0.04t} = 45$ for t in terms of natural logarithms, and find its value rounded to two decimal places.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angle bisectors and perpendicular bisectors, $m\angle A = 41^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 94°

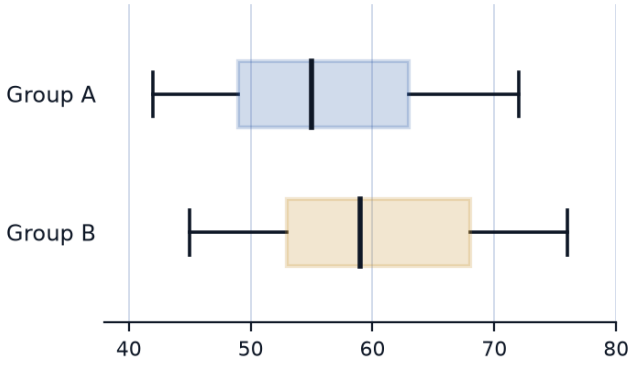
b. 104°

c. 86°

d. 139°

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying altitudes, medians, and bisectors of triangles, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 93°

b. 103°

c. 85°

d. 138°

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 9$ in dividing a segment into a given ratio, evaluate $f(4)$.

a. 21

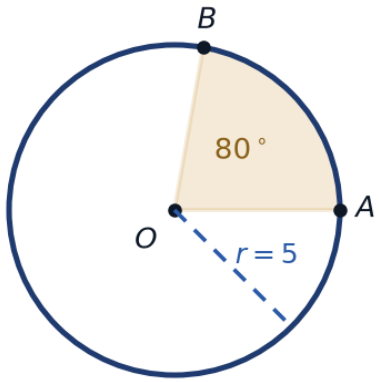
b. 24

c. 12

d. 25

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two chords intersect inside a circle. One chord is divided into segments of length 5 and 8; the other into segments of length 10 and x. Find x.



a. 10

b. 4

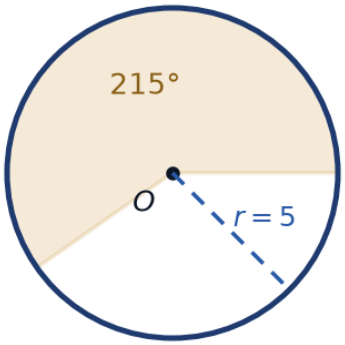
c. 6

d. 13

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A central angle in a circle measures 215°. What is the measure of its intercepted arc?



a. 35°

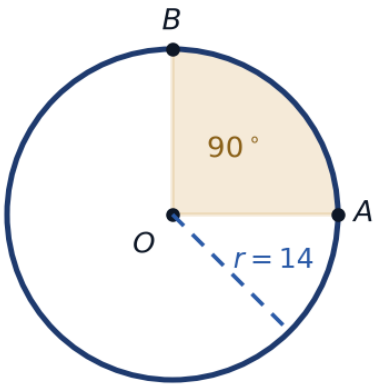
b. 107°

c. 145°

d. 215°

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An inscribed angle intercepts an arc of 144°. What is the measure of the inscribed angle?



a. 144°

b. 108°

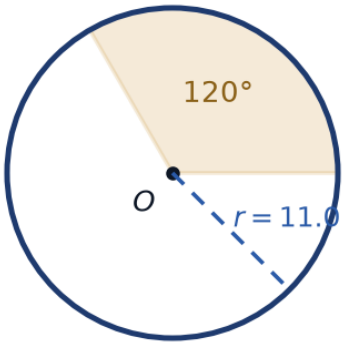
c. 216°

d. 72°

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circle has radius 11. Find the area of a sector with central angle 120°, in terms of π.



a. $\frac{121}{3}\pi$

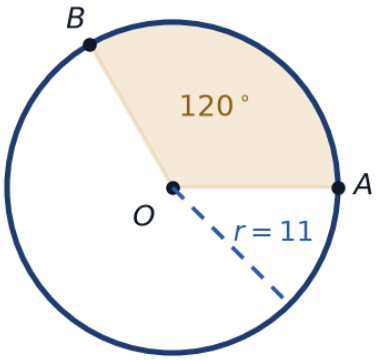
b. $\frac{122}{3}\pi$

c. 121π

d. $\frac{123}{3}\pi$

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A circle has radius 11. Find the arc length of a sector with central angle 120°, in terms of π.



a. 22π

b. 11π

c. $\frac{23}{3}\pi$

d. $\frac{22}{3}\pi$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In the xy-plane, the circle given by the equation $x^2 + y^2 - 4x + 6y - 12 = 0$ is tangent to line ℓ at the point $(5, 1)$. Which of the following is an equation of line ℓ?

a. $3x + 4y = 19$

b. $4x + 3y = 23$

c. $3x - 4y = 11$

d. $4x - 3y = 17$

30 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

In the circle shown, chords AB and CD intersect at point P. If $PA = 6$, $PB = 8$, and $PC = 4$, what is the value of x, the length of segment PD?

a. 10

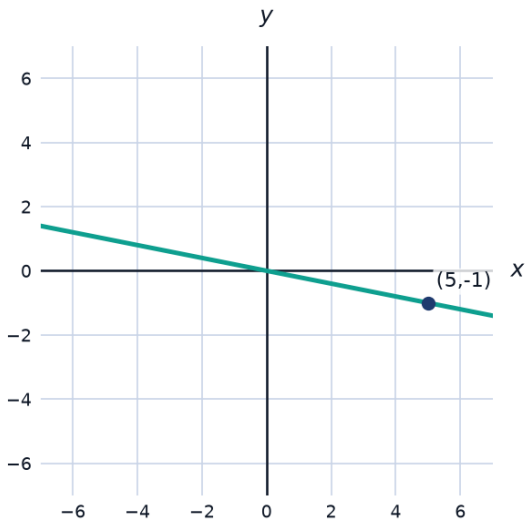
b. 12

c. 16

d. 18

31 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Point $(5, -1)$ is dilated about the origin with scale factor 2. What are the coordinates of the image?



a. (10, -2)

b. (10, -1)

c. (7, 1)

d. (-10, 2)

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32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A rectangle has length 7 and width 6. What is its perimeter?



a. 30

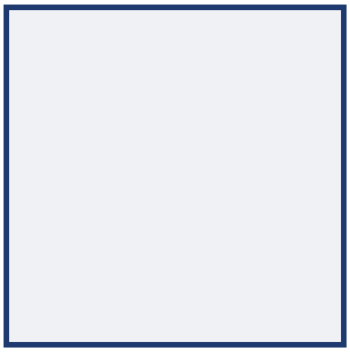
b. 26

c. 13

d. 42

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A regular 4-gon has side length 2 and apothem 11. What is its area?



a. 88

b. 44

c. 8

d. 50

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A test consists of 25 multiple-choice questions. A student scored 80%. If the student correctly answered 4 more questions, what would the new percentage score be?

a. 84%

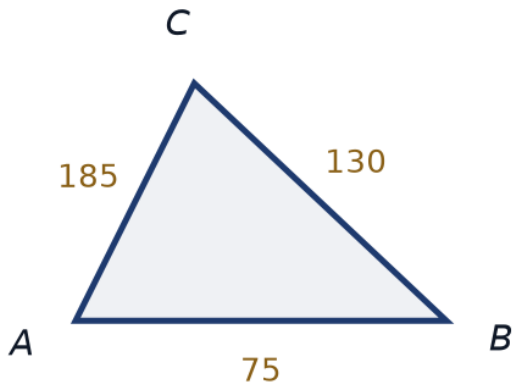
b. 88%

c. 92%

d. 96%

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A triangle has sides 75, 130, and 185. Using Heron's formula, find its area.



a. 2240

b. 2250

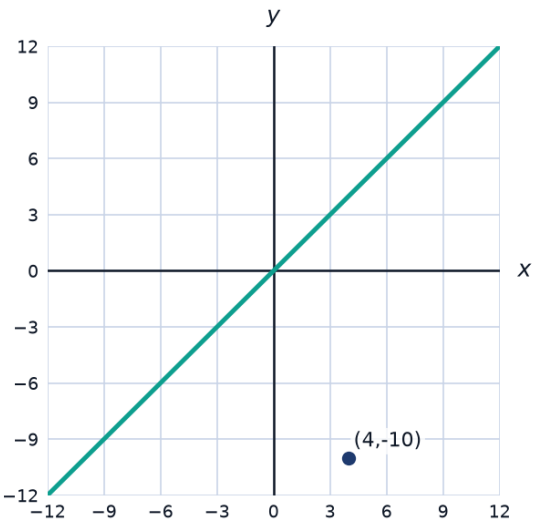
c. 390

d. 2265

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Point $(4, -10)$ is reflected over the line $y = x$. What are the coordinates of the image?



a. (-4, -10)

b. (10, -4)

c. (-10, 4)

d. (4, -10)

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A rectangular prism has length 3, width 3, and height 10. What is its surface area?



a. 16

b. 146

c. 90

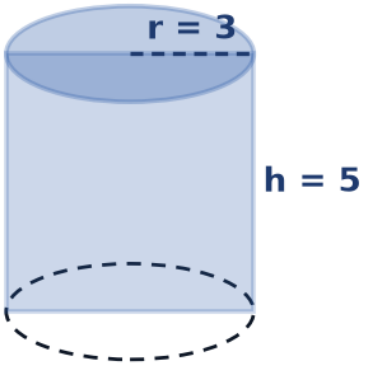
d. 138

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Total surface area of a cylinder with radius 3 and height 5 (in terms of π):



a. 57π

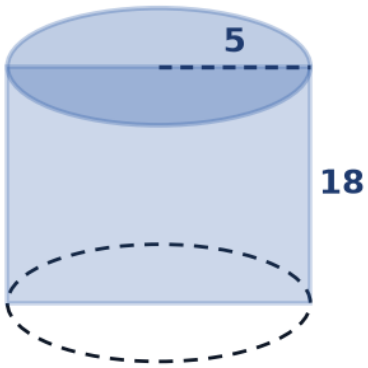
b. 30π

c. 15π

d. 48π

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A cylinder has radius 5 and height 18. Find its total surface area in terms of π.



a. 180π

b. 450π

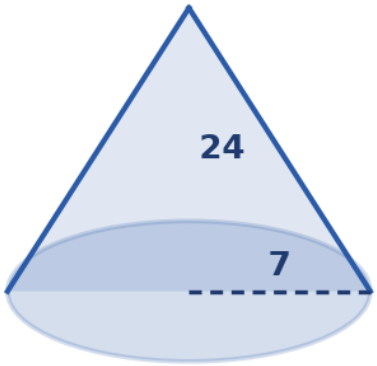
c. 239π

d. 230π

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A cone has radius 7 and height 24. Find its volume in terms of π.



a. 168π

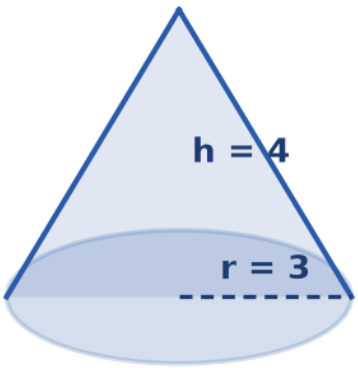
b. 397π

c. 392π

d. 1176π

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Area, Surface Area & Volume

Volume of a cone with radius 3 and height 4 (in terms of π):



a. 36π

b. 12π

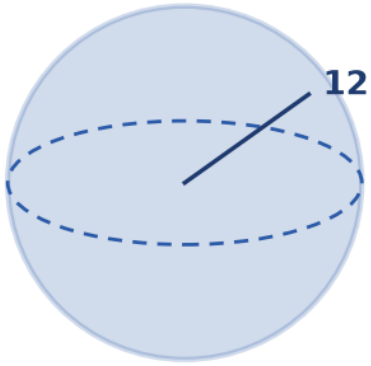
c. 19π

d. 4π

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Transformations & Symmetry

A sphere has radius 12. Find its volume in terms of π.



a. 1728π

b. 2311π

c. 576π

d. 2304π

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Foundations & Reasoning

A trapezoid has parallel sides 4 and 11 and height 20. What is its area?



a. 35

b. 162

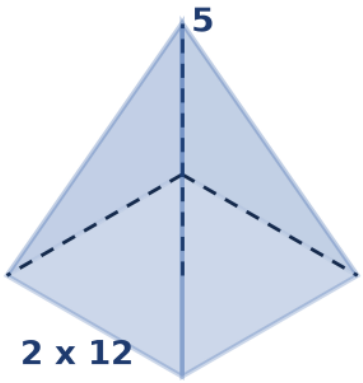
c. 300

d. 150

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Coordinate Geometry

A rectangular pyramid has a base of length 2 and width 12, and a height of 5. What is its volume?



a. 120

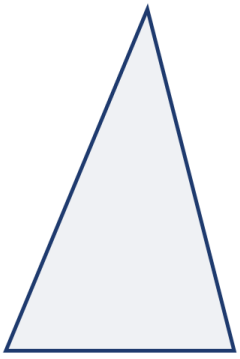
b. 48

c. 40

d. 19

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Probability

A triangle has base 6 and height 9. What is its area?



a. 30

b. 54

c. 27

d. 15

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